REV 2021 SEMESTER 2 CIVIL ENGINEERING

Basic Surveying Lab

Course Code: 2019 | Credits: Nil

Course Overview

Category	Engineering Science	**Total Hours**	45 Periods
Periods/Week	3 (Practical)	**Credits**	Nil

This course provides hands-on training in fundamental surveying techniques. Students learn to use chains, tapes, cross-staffs, prismatic compasses, and levelling instruments to measure distances, angles, and elevations for engineering projects.

Course Outcomes (CO)

CO No.	Description	Duration	Cognitive Level
CO1	Plot land area using principles of chain survey and plane table survey	10 Hours	Applying
CO2	Plot land area by compass survey	12 Hours	Applying
CO3	Determine level difference between stations in the field	9 Hours	Applying

CO No.	Description	Duration	Cognitive Level
CO4	Sketch longitudinal and cross profiles	10 Hours	Applying

Module 1: Chain Survey & Plane Table Survey

10 Hours CO1 Coverage

M1.01: Distance Measurement (Tape)

Measuring the distance between two intervisible stations using a tape is the most basic surveying operation. It involves ranging (aligning intermediate points) and chaining.

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- **Direct Ranging:** Used when stations are intervisible.
- **Indirect Ranging:** Used when high ground or obstacles prevent seeing one station from another.

M1.02: Cross Staff Survey

The cross-staff is used to set out perpendicular offsets from a main survey line to boundary points. The area is then calculated by dividing the field into triangles and trapezoids.

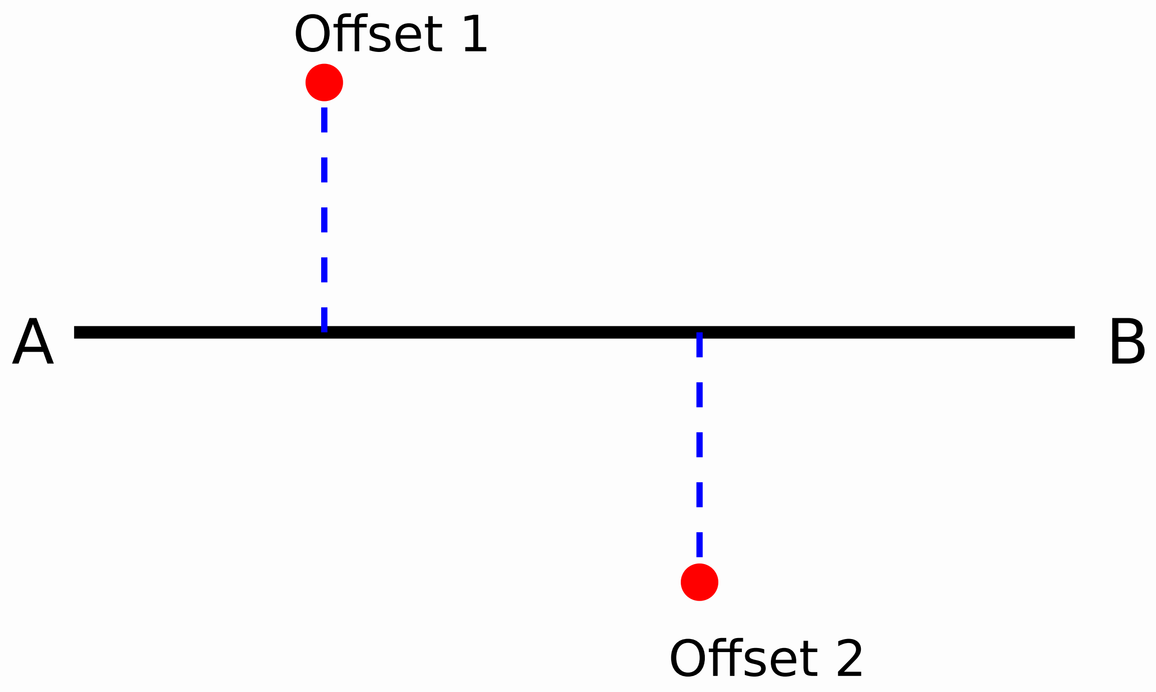

Figure 1: Setting offsets using Cross Staff along Chain Line AB.

M1.03: Plane Table Surveying

Plane table surveying is a graphical method where field observations and plotting are done simultaneously.

- **Radiation Method:** Used when all points to be plotted are visible and accessible from a single station.
- **Intersection Method:** Used for inaccessible points. The point is sighted from two different stations, and the intersection of the rays gives the position.

Module 2: Compass Surveying

12 Hours CO2 Coverage

M2.01 & M2.02: Compass Principles

The Prismatic Compass measures the magnetic bearing of lines. The **Whole Circle Bearing (WCB)** system is used, ranging from 0° to 360°.

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Calculation of Included Angles:
Included Angle = Bearing of Next Line - Bearing of Previous Line.
If the result is negative, add 360°.

M2.03: Closed Traverse

A closed traverse starts and ends at the same point (or at points whose positions are known). For a 5-sided traverse, the sum of internal angles should be `(2n - 4) * 90°`.

Sum of Interior Angles = $(n - 2) \times 180°$

Module 3: Levelling

9 Hours CO3 Coverage

M3.01: Simple Levelling

Levelling is the operation to determine the relative heights of points on the earth's surface. **Simple Levelling** is used to find the difference in elevation between two points visible from a single instrument station.

M3.02: Reduction of Levels

There are two main methods to calculate Reduced Levels (RL):

1. Height of Instrument (HI) Method

HI = RL of BM + Back Sight (BS)

RL of Point = HI - FS (or IS)

2. Rise and Fall Method

Difference = Previous Reading - Current Reading

Positive = Rise; Negative = Fall

New RL = Old RL + Rise (or - Fall)

Quick RL Calculator (HI Method)

Benchmark RL (m): 100.000

Back Sight (BS) (m): 1.250

Fore Sight (FS) (m): 2.100

Calculate RL

Module 4: Profiling (L-Section & C-Section)

10 Hours CO4 Coverage

M4.01: Fly Levelling

Fly levelling is performed to determine the RL of a benchmark at a distance from the starting point. It involves only Back Sights and Fore Sights.

M4.02: Longitudinal & Cross Profiles

Used extensively in road, rail, and canal projects.

- **Longitudinal Section (L-Section):** Levels taken along the centerline of the proposed alignment to determine the gradient.
- **Cross Section (C-Section):** Levels taken perpendicular to the centerline to determine the earthwork volume.

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Formula Bank

Area of Triangle (Chain Survey)

Area = ½ × Base × Height

Included Angle (Compass)

FB of next line - BB of prev line

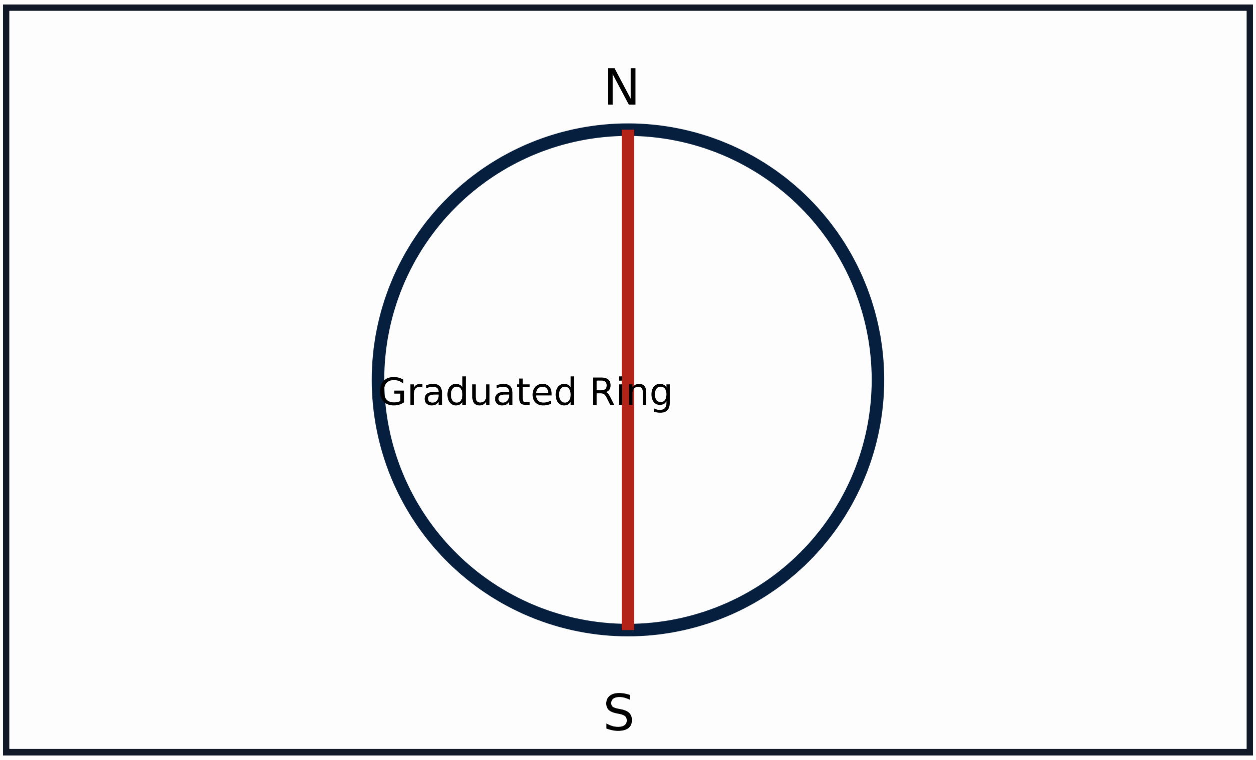
HI Method Check

ΣBS - ΣFS = Last RL - First RL

Rise & Fall Check

ΣRise - ΣFall = Last RL - First RL

Visual Library



Simplified Prismatic Compass Schematic

Module-wise Practice Questions

Q1: What is the purpose of 'Ranging' in chain survey? (3 Marks)

Ranging is the process of establishing intermediate points on a straight line between two survey stations. It ensures that the distance measured with the tape is along a straight line, which is the shortest distance.

Q2: Differentiate between Radiation and Intersection methods in Plane Table Survey. (5 Marks)

Radiation: Points are located by drawing rays from a single station. Distances are measured on ground and plotted to scale. Suitable for small, accessible areas.
Intersection: Points are located by the intersection of rays drawn from two different stations. No ground distance measurement to the points is needed. Suitable for inaccessible points like a point across a river.

Q3: Define Fore Bearing and Back Bearing. (2 Marks)

Fore Bearing (FB) is the bearing of a line in the direction of the progress of the survey. Back Bearing (BB) is the bearing of the line in the direction opposite to the progress of the survey. BB = FB ± 180°.

Practice Model Practical Paper

End Semester Practical Examination

Time: 3 Hours | Max Marks: 100

1. Conduct a closed compass traverse for the given 5 stations. Calculate the included angles and check for local attraction. (40 Marks)
2. Find the difference in level between two given points A and B using a Dumpy Level. Perform the arithmetic check. (30 Marks)
3. Plot the boundary of the given field using the Cross-Staff survey method. (20 Marks)
4. Viva Voce and Record. (10 Marks)

Quick Revision

- **Chain Survey:** Principle is Triangulation.
- **Compass Survey:** Principle is Traversing.
- **Plane Table:** Graphical method; no field book required.
- **Levelling:** BS is the first reading; FS is the last reading before shifting.
- **Local Attraction:** Disturbance to the compass needle by magnetic objects. Detected if FB ~ BB difference is not 180°.

Glossary

Benchmark (BM)
: A fixed point of known elevation above a datum.

Reduced Level (RL)
: The vertical distance of a point above or below the datum.

Offset
: Lateral distance measured from the main survey line to a feature.

Official Source

This content is based on the official Kerala Polytechnic Revision 2021 syllabus for Course Code 2019 (Basic Surveying Lab) provided by SITTTR Kerala.

[Official SITTTR Source]

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